

Emergency Medicine in Practice

IB 198-002 Fall 2025

2 unit P/NP|| Enrollment Permission Required

1 section: Monday 6-8 p.m.

Facilitators: Eva Reineck and Fontanna Zhi, OH by appointment

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Faculty Sponsor: Caroline Williams (cmw@berkeley.edu)

Course Description

This course is designed to emphasize the application of basic emergency medical skills in a collaborative group work setting. Students will engage with the field of emergency medicine by reading current events and research at the level of an emergency medical technician (EMT) and will present their findings to the class in a student-led format. In addition to a knowledge-based component, students will have the opportunity to use medical equipment on simulated patients through scenarios, which will mimic the workflow and energy of a real-life emergency situation, and receive constructive feedback to improve both technical and communication skills. Students will be provided opportunities to volunteer or work at actual events as medical providers or assistants, but, if unable to attend these, will also gain knowledge regarding working in emergency medicine by interacting with multiple guest speakers we will host throughout the semester.

Learning Objectives

After taking this course, students will:

- Practice applying EMT-basic level skills in scenarios, including becoming familiar with and utilizing medical equipment and transportation devices
- Develop confidence with patient communication skills through consistent practice and constructive feedback from fellow students
- Improve teaching skills by working collaboratively to research and create informal and engaging presentations to peers
- Gain experience applying skills learned in the classroom in field opportunities provided by the facilitators (volunteering at EMT events, working on an ambulance, teaching public health classes, etc.)
- Understand the role of EMS in the larger scope of the medical field by engaging with guest speakers working in EMS

Enrollment Prerequisites

Interested students must complete a google form application that will include questions about their current level of medical certification. We highly recommend you have an active EMT license and certification in the state of California, since we will plan our lecture sessions around

an assumed knowledge at the EMT-basic level, but this course is open to anyone with a demonstrated interest and dedication to engaging with emergency medicine at this level, regardless of previous certification. Permission from the instructors will be required to enroll.

Course Structure

Each week, we will cover a different topic in emergency medicine as outlined in the course syllabus, or we will host a guest speaker who will provide their own topic to discuss in advance. Class meetings will go in-depth on the week's topic in a student-led format described in the "Assignments" section below.

Attendance: Class meetings will be held in one section on Monday from 6-8pm. Synchronous, in-person attendance at this section each week is required for all meetings unless otherwise stated by the facilitators in advance. Absences will be excused under limited circumstances (i.e. illness, personal/family emergency, etc.). It is the responsibility of the student to inform the instructor of such circumstances prior to the absence, if possible. Students are allowed a maximum of 2 unexcused absences throughout the semester. During these 2 unexcused absences, we hope to provide a link to a Zoom recording of the presentation for that week along with a makeup assignment that can be completed.

Readings: Brief, 1-2 page primers of the material will be provided as a PDF to all students in advance of the class meeting on our bCourses site as indicated by the course schedule. It is highly recommended that students review this information as the lecture presentation and scenario quiz will be based on it. Where the course calendar says readings are TBD or there is a guest speaker, readings to review will be provided based on the guest speaker's or the presenter's topic at least a week in advance of the class meetings. These could take the form of news articles, research papers, or 1-2 page primers made by the presenting teams.

Assignments: Beginning from the second class, students will pick a topic they are interested in based on the course schedule and sign themselves up for that topic on a google sheet. Groups of 2-5 students who selected that topic will then gather relevant research and news on the topic in the field of EMS and present their findings to the class in their format of choice (a combination of slides, worksheets, scenarios, etc.) with the aid of the facilitators. The presentation is expected to last around 40 minutes to 1 hour with interactive components to engage other classmates. It is also required that the students prepare 3 patient-case scenarios relevant to their topic which engage other students on their EMT-level knowledge and skills. The presenting students are expected to start researching and drafting their presentation around two weeks before the presentation date and submit their slides to facilitators for revision before their presentation. Students are expected to meet outside of class to plan, organize, and rehearse their presentations. From previous experience, the preparation work for the presentation takes on average 8 hours. Students who are not presenting will be asked to engage with the lesson by writing down notes

and questions. After class, students will be required to fill out a written assignment through bCourses, which will ask them to write down three new pieces of information they have learned or questions they still have, which will be addressed at the beginning of the following class. Additionally, we will provide two google forms to obtain feedback from students throughout the semester and adapt our course content and structure accordingly to ensure students feel supported and comfortable with the difficulty level of the material. There will also be a final reflection assignment about the course overall.

Scenarios: Complementary to the topic we or the guest speaker are covering for the week, we will post a bCourses quiz with various scenarios for students to apply their knowledge after they have reviewed the primer and listened to the class presentation. These scenarios will provide an opportunity for students to consider real-world situations and what the best courses of action would be based on what they have learned. After the quiz is closed, the full scenario and recommended interventions will be provided to the students and the facilitators will review the most common errors in the following week's meeting.

Mass Casualty Incident (MCI) Drill: As part of our dedication to emergency service work and disaster preparedness, students are required to participate in a mass casualty incident drill during the last week of instruction. This drill is designed to integrate clinical skills, rapid decision-making, and coordinated team response and encourage participants to apply principles such as START triage, resource allocation, and crisis communication. Students will be asked to write a reflection on this experience. If unable to participate, students can discuss other options to practice these skills with the facilitators.

Additional Time Commitments: Students will be provided with various methods to engage in the EMS field outside the classroom, regardless of their level of experience. These opportunities range from working at university sports events, staffing concerts, ride-alongs on ambulances, teaching public health courses, staffing UHS clinics, and others. This will not be a hard requirement, due to varying levels of medical certification and time commitments, but it is highly encouraged students take advantage of these opportunities. Students will be asked to write a reflection on this experience. If unable to participate, students can discuss other options to practice their skills with the facilitators.

Course Calendar

Below is a tentative outline of course topics and scheduled dates for guest speakers. Depending on guest speaker availability, this timeline may change slightly. Also, depending on student interests, we may change some of the topics later in the semester to reflect what students would like to learn in the class.

date	lesson topic
2/2	new member testing
2/7	new member academy
2/9	cardiac emergencies
2/16	FA/STB instructor training
2/23	respiratory emergencies
3/2	trauma care with emphasis on STB, spinal immobilization, and splinting
3/9	EKG training
3/16	environmental emergencies
3/23	spring break
3/30	guest speaker
4/6	psychiatric and behavioral emergencies
4/13	pediatrics and obstetrics
4/20	toxicology - poisoning, overdose, drugs
4/27	disaster training and mass casualty incident preparation
5/2	MCI

We will attempt to record and upload all sessions to the Media Gallery on bCourses.

Reading List

All chapters refer to '*Emergency Care and Transportation of the Sick and Injured 12th Edition*'

- Pollak, A. N., Mejia, A., McKenna, K., & Edgerly, D. (2021). Emergency care and transportation of the sick and injured. Jones & Bartlett Learning.

9/12: FTO Training + New Member Academy

- [2025 Alameda County EMS Field Manual](#)

9/15: Cardiac Emergencies

- Chapter 12: Pharmacology (focus on cardiac-related medications)
- Chapter 13: Shock
- Chapter 14: BLS Resuscitation
- Chapter 17: Cardiovascular Emergencies

9/22: CPR and STB/FA Instructor Training

- Chapter 14: BLS Resuscitation
- Chapter 16: Respiratory Emergencies

9/29: Respiratory Emergencies

- Chapter 11: Airway Management

- Chapter 13: Shock
 - Chapter 14: BLS Resuscitation
 - Chapter 16: Respiratory Emergencies
- 10/6: Trauma Care
- Chapter 25: Trauma Overview
 - Chapter 26: Bleeding and Shock
 - Chapter 27: Soft Tissue Trauma
 - Chapter 28: Face and Neck Injuries
 - Chapter 29: Head and Spine Injuries
 - Chapter 30: Chest Injuries
 - Chapter 31: Abdominal Injuries
 - Chapter 32: Orthopedic Injuries
- 10/13: Special Training Event
- Reading TBD
- 10/20: Environmental Emergencies
- Chapter 13: Shock (as applicable to environmental causes)
 - Chapter 33: Environmental Emergencies
- 10/27: Guest Speakers
- Reading TBD
- 11/3: Psychological and Behavioral Emergencies
- Chapter 3: Medical, Legal, and Ethical Issues
 - Chapter 23: Behavioral Emergencies
- 11/10: Toxicology – Poisoning, Overdose, Drugs
- Chapter 12: Pharmacology
 - Chapter 15: Medical Overview
 - Chapter 22: Poisoning and Overdose Emergencies
- 11/17: Pediatrics and Obstetrics
- Chapter 7: Lifespan Development
 - Chapter 24: Gynecologic Emergencies
 - Chapter 34: Obstetric and Neonatal Emergencies
 - Chapter 35: Pediatric Emergencies
- 12/1: Disaster and MCI Preparedness
- Chapter 38: Transport Operations
 - Chapter 39: Vehicle Extrication and Special Rescue
 - Chapter 40: Incident Management
 - Chapter 41: Terrorism Response

Grade Breakdown

This course is P/NP only. Grades will be given according to the breakdown below:

- Attendance - 50%
- Student-led lecture preparation - 20%
- Outside of classroom EMS engagement or alternative assignment - 15%
- Post-class writeup on bCourses - 5%
- Scenario weekly quizzes - 5%
- Two Google Feedback forms - 2.5%
- Final reflection on course (required) - 2.5%

We will post updated grade information on our bCourses site after each week. Students must receive a score of 70% AND complete the final reflection to pass.

Students with Disabilities

Students who require accommodations due to a physical, emotional, or learning disability should contact UC Berkeley's Disabled Students' Program (DSP). Please notify the instructor of any accommodation requests made for this course.

Resources

These resources will be linked in our bCourses under “Files” and will be sent out on a weekly basis based on the topic of the week. Refreshers are required for certain topics as indicated by the course outline above, and we encourage you to explore the other provided resources and your own for further study on the topic (even if you are not leading the presentation that week!)

- Refreshers referenced in the course outline above are linked below:
 - [Cardiac Emergencies](#)
 - [Trauma Care](#) and [Stop The Bleed slides](#)
 - [Environmental Emergencies](#)
 - [Pediatric Care](#)
 - [Disaster Training](#)
- [Alameda County Protocols](#) to review for the first meeting and reference throughout the semester for different topics
 - Additionally, there is an [ALCO EMS app](#) for quick reference on your cell phone
- [Google Drive folder](#) with multiple examples of Patient Care Reports (PCRs) of example scenarios - EMTs are required to fill this paperwork out after medical interventions
- EMS websites with research and news
 - Annals of Emergency Medicine: <https://www.annemergmed.com/>
 - Emergency Medicine Journal: <https://emj.bmj.com/>
 - EMS News: <https://www.ems1.com/ems-news/>
 - EMS World: <https://www.hmpgloballearningnetwork.com/site/emsworld/archive>
 - Journal of Emergency Medical Services: <https://www.jems.com/>
 - NREMT resources: <https://www.nremt.org/Document/resources>